

Member ID: \_\_\_\_\_

Time: \_\_\_\_\_

Rank: \_\_\_\_\_



# ADVANCED SPREADSHEET APPLICATIONS (235)

## STATE 2024

### PRODUCTION

Job 1: Format Spreadsheet/Queries/Advanced Functions \_\_\_\_\_ (125 points)

Job 2: Subtotaled List and Advanced Functions \_\_\_\_\_ (75 points)

Job 3: Create and Modify Chart \_\_\_\_\_ (75 points)

*TOTAL POINTS* \_\_\_\_\_ (275 points)

**Test Time: 90 minutes**

## GENERAL GUIDELINES:

*Failure to adhere to any of the following rules will result in disqualification:*

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

## EXAM GUIDELINES:

1. Ensure this test booklet contains Jobs 1-3.
2. Key all jobs according to the instructions given.
3. Correct any and all formatting, spelling or grammar errors. Use the formatting guide in the *Style & Reference Manual*.
4. Your name or initials should *not* appear on any work you submit. Use your Member ID in any occasion you would normally key your reference initials.
5. In the lower **right** hand corner of ALL work submitted (unless otherwise specified), key your Member ID and job number.
6. If you complete the event before the end of the time allotted, notify the proctor. Time may be considered a factor in determining a winner when there is a tie score.
7. Place your scoring sheet on top of your jobs. Jobs should be placed in numerical order.

| <b>Job 1 (Format Spreadsheet/Queries/Advanced Functions)</b>                                                                                                       | <b>Points Possible</b> | <b>Points Earned</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------|
| Member Number and Job Number in right section of footer                                                                                                            | 5                      |                      |
| Format Cell A1 with Heading 1 cell style                                                                                                                           | 5                      |                      |
| Merge Cell A1 across columns A-G                                                                                                                                   | 5                      |                      |
| Format data in column A with date format dd-mmm (01-Jan)                                                                                                           | 5                      |                      |
| Format Cells A3-D70 as a table, using Green, Table Style Dark 11 (should have drop-down lists in column headings)                                                  | 10                     |                      |
| Using Queries feature, create a query, bringing in data from provided text file.                                                                                   | 10                     |                      |
| Rename columns in Sheet2 to City, County, and Avg Home Price                                                                                                       | 5                      |                      |
| In Sheet1, insert two columns between columns C and D. Name new columns "County" and "Avg Home Price"                                                              | 10                     |                      |
| Column D has VLOOKUP function to insert county from Sheet 2, column D                                                                                              | 10                     |                      |
| Column E has VLOOKUP function to insert average home price from Sheet 2 into Column E                                                                              | 10                     |                      |
| All 6 instances of Sioux Falls in column C is linked to webpage <a href="https://www.siouxfalls.org">https://www.siouxfalls.org</a>                                | 10                     |                      |
| Add column heading to column G, "Price Compared to Avg Home Price," and insert formula B4-E4 to cells in column G                                                  | 10                     |                      |
| Use arrow icons in column G to indicate a green "up" arrow for any cell that is greater than or equal to 0. Cells that are less than 0 should not show any arrows. | 10                     |                      |
| Print data in portrait orientation, fitting all columns to one page                                                                                                | 10                     |                      |
| Hide all columns except column D, and print again, showing formulas, in landscape, resizing column D so that entire VLOOKUP function is visible                    | 10                     |                      |
| <b>Subtotal</b>                                                                                                                                                    | <b>125</b>             |                      |

| <b>Job 2 (Subtotaled List and Advanced Functions)</b>                                                                  | <b>Points Possible</b> | <b>Points Earned</b> |
|------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------|
| Member Number and Job Number in right section of footer                                                                | 5                      |                      |
| Unhide all columns from Job 1 spreadsheet                                                                              | 5                      |                      |
| Sort data by County (ascending), then by Property Location (ascending), and then by Listing Price (descending)         | 15                     |                      |
| Convert the table to a range (it will no longer have the drop-down list arrows in column headings)                     | 10                     |                      |
| Use Subtotal feature to group and calculate the average Listing Price by County. Include Grand Average at bottom       | 20                     |                      |
| Format font for subtotaled averages and grand average in Accounting format, with a dollar sign, 0 decimal places, bold | 15                     |                      |
| Print worksheet in portrait orientation showing data, scaling to fit all columns on one page                           | 5                      |                      |
| <b>Subtotal</b>                                                                                                        | <b>75</b>              |                      |

| <b>Job 3 (Create and Modify Chart)</b>                                                                      | <b>Points Possible</b> | <b>Points Earned</b> |
|-------------------------------------------------------------------------------------------------------------|------------------------|----------------------|
| Member number and job number in right section of footer                                                     | 5                      |                      |
| Filter data to select only six cities (Aberdeen, Germantown, Harrisburg, Mitchell, Rapid City, Sioux Falls) | 10                     |                      |
| Create 3D Clustered Bar chart, using Style 3, using County and Avg Home Price                               | 15                     |                      |
| Place chart on separate sheet in workbook, resize to fit Cells A1-K22                                       | 10                     |                      |
| Change chart title to AVERAGE HOME PRICE BY COUNTY                                                          | 5                      |                      |
| Format text outline to solid line, 0.5 width (will affect legend labels)                                    | 10                     |                      |
| Change chart style to 5, showing data across bottom of chart                                                | 10                     |                      |
| Print chart in landscape orientation, centering horizontally and vertically on page                         | 10                     |                      |
| <b>Subtotal</b>                                                                                             | <b>75</b>              |                      |
| <b>TOTAL POINTS</b>                                                                                         | <b>275</b>             |                      |

**As an employee of Precision Realty Company, you have been asked to format and analyze data and prepare reports and charts.**

**JOB 1: Format Spreadsheet/Queries/Advanced Functions**

1. Open *Job1Data* file that was provided to you.
2. Format Cell A1 with Heading 1 cell style, and merge and center the title across Columns A-G.
3. Format the data in column A with the date format dd-mmm (01-Jan).
4. Format Cells A3-D70 as a table, using Green, Table Style Dark 11.
5. Using the Queries feature, create a Query, getting data from the text/csv file named *City-CountyData* text file that was provided to you, and load it to a new worksheet (within the Job1Data file workbook you currently have open).
6. Rename the columns in Sheet2 “City” for A1, “County” for A2, and “Avg Home Price” for A3.
7. In Sheet1, insert two new columns between Columns C and D. Name the first new column “County” and the second new column “Avg Home Price.”
8. In Cell D4, use the VLOOKUP function to insert the county from Sheet2 in Column D. Ensure that the formula has populated all of column D.
9. In Cell E4, use the VLOOKUP function to insert the average home price from Sheet 2 in Column E. Ensure that the formula has populated all of column E.
10. In Sheet1, create a link in Cell C4 to the webpage <https://www.siouxfalls.org/>. Create links to the same page to all six instances of Sioux Falls in Column C.,
11. In Column G, type column heading “Price Compared to Avg Home Price,” and wrap text, and then insert a formula in the column which will subtract the Avg Home Price in column E from the Listing Price in column B.
12. Format the cells in Column G with conditional formatting, using the arrow icons to indicate a green “up” arrow for any cell in Column G that is greater than or equal to 0. Cells that are less than 0 should not have any icon. Format the data in column G with the comma style, no decimals.
13. Add your Member ID# and Job 1 in right footer. Auto-fit all columns widths.
14. Print spreadsheet showing data in portrait orientation, fitting all columns to one page.
15. Hide all columns except Column D, and print worksheet showing formulas in landscape orientation, resizing Column D so that entire VLOOKUP function is visible (check your print preview to ensure function is not truncated).

## **JOB 2: Subtotaled List and Advanced Functions**

1. Using the spreadsheet from Job 1, unhide all columns to show data.
2. Sort the data in the sheet first by County in ascending order, then by Property Location in ascending order, and then by Listing Price, largest to smallest.
3. Convert the table in this worksheet (Sheet 1) to a range.
4. Subtotal the data by County, calculating the average of Listing Price for each county. Include a Grand Average at the bottom.
5. Format the subtotaled averages by County in Accounting format, with a dollar sign and 0 decimal places, and bold each of those averages. Format the Grand Average in the same manner.
6. Add Member ID# and Job 2 in right footer.
7. Print worksheet in portrait orientation, showing data, scaling to fit all columns on one page.

## **JOB 3: Create and Modify Chart**

1. Open Sheet 2 from the earlier jobs.
2. Filter data to select only the cities of Aberdeen, Germantown, Harrisburg, Mitchell, Rapid City, and Sioux Falls.
3. Select the county and avg home price columns and create a 3D Clustered Bar chart, using Style 3.
4. Create a new worksheet. Place the chart on this sheet in the workbook and resize it to fit cells A1 through K22.
5. Change the title to AVERAGE HOME PRICE BY COUNTY.
6. Use Chart Styles to change the chart style to 5, showing the data across the bottom of the chart.
7. Add a data table. Using Text Options, format the Text Outline to solid line, with a 0.5 width on the data table.
8. Add Member ID and Job 3 in right footer.
9. Print chart in landscape orientation, centering chart horizontally and vertically on page.